

CO3117 - Release Baseline diagnostic (W01 - W04)

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Date: 25/9/2025

Start: 10:00 - End: 11:20

Reference: - Tom M Mitchell, Machine Learning (1997): Q1 (Ex 3.2), Q2 (Ex 3.1), Q7 (Ex 13), Q3-5 are generated by AI based on 37.1
- A. Muller & S. Guido, Introduction to ML with Python (2017).
Q6 is generated by AI based on chapter 2, pages 26-29 and 56 of this book

Questions:

Q1:

Instance	Classification	a_1	a_2
1	+	T	T
2	+	T	T
3	-	T	F
4	+	F	F
5	-	F	T
6	-	F	T

a) What is the entropy of this collection of training examples with respect to the target function classification?
b) What is the information gain of a_2 relative to these training examples

Q2: Give decision trees to represent the following boolean functions

a) $A \wedge B$

c) $A \text{ XOR } B$

b) $A \vee [B \wedge C]$

d) $[A \wedge B] \vee [C \wedge D]$

Q3: What is overfitting? Sketch the accuracy on training set and test set as function of decision-tree size. Mark the

region where the tree starts to overfit.

Q4: Name 2 approaches to avoid overfitting in decision tree learning. Describe how reduced-error pruning works.

Q5: a) How does a decision tree handle a continuous-valued attribute?

b) How can a training example with missing attribute value be handled?

Q6: a) As model complexity increases, how do the training accuracy and test accuracy change? Mark the underfitting and overfitting regions.

b) How does logistic regression differ from linear regression? Why can logistic regression be used for classification?

- Answer:

Q1: a) We have: Entropy = $-\sum p(x) \cdot \log(p(x))$

$$\begin{aligned} &= - [p(+) \cdot \log_2 p(+) + p(-) \cdot \log_2 p(-)] \\ &= - \left[\frac{1}{2} \cdot \log_2 \left(\frac{1}{2} \right) + \frac{1}{2} \cdot \log_2 \left(\frac{1}{2} \right) \right] \\ &= 1 \end{aligned}$$

Information gain of a_2 : $\text{Gain}(\text{Entropy}, a_2) = \text{Entropy}(S) - \sum_k \frac{|S_k|}{|S|} \cdot \text{Entropy}(S_k)$

Calculate Entropy (S_K) ($K = \{T, F\}$)

For $K=T$

$$\text{Entropy}(S_T) = -\left(\frac{1}{2} \log_2 \frac{1}{2} + \frac{1}{2} \log_2 \frac{1}{2}\right) = 1$$

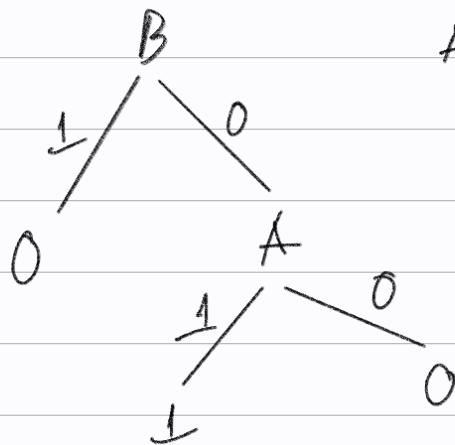
$K=F$

$$\text{Entropy}(S_F) = -\left(\frac{1}{2} \log_2 \frac{1}{2} + \frac{1}{2} \log_2 \frac{1}{2}\right) = 1$$

$$\Rightarrow \text{Gain}(S, a) = 1 - \left(\frac{4}{6} \cdot 1 + \frac{2}{6} \cdot 1\right) = 0$$

Q2

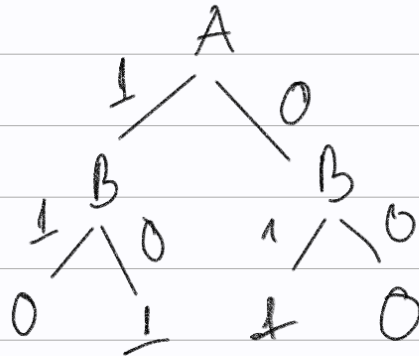
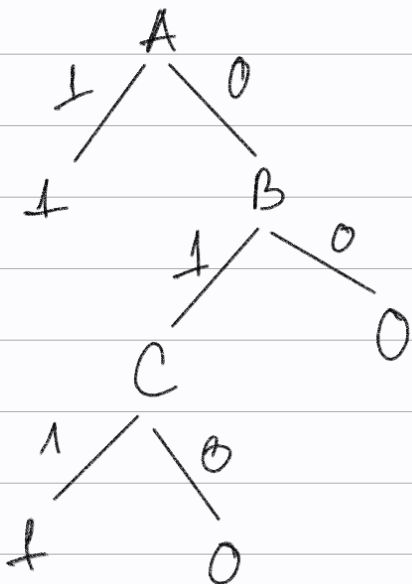
a)



$A \wedge \neg B$

b) $A \vee [B \wedge C]$

c) $A \text{ XOR } B$



- The reduced-error pruning is technique replaces leaf which enhances accuracy

Q₅

- a) To handle continuous values, we use regression tree
- b) Don't Know

Q₆ : a) Complexity increases $\rightarrow$ training accuracy increases
testing accuracy decreases

b) Linear regression is the technique that predicts based on linear function

Logistic Regression is technique of predicting based on non-linear function (sigmoid)

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